RESEARCH MEETING / PAPER SUMMARY

HBF Sucks?

A Full-Stack Characterization of High-Bandwidth Flash for KV-Centric LLM Serving

A faster flash tier can make the serving system slower.

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National Taiwan University | Academia Sinica | 27 Oct 2026

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arXiv:2608.11668v4

Why Replace SSD with HBF?

MOTIVATION / THE INTUITIVE UPGRADE

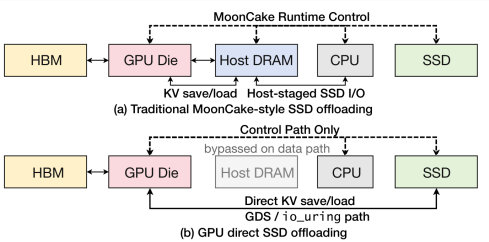


Figure 3: SSD offloading of KV: host-staged path (a) versus GPU-direct path (b).

KV cache outgrows GPU memory
Save / load KV to an SSD backing tier, through host staging or GPU-direct I/O.

HBF promises faster flash
Package-local flash offers lower latency and higher bandwidth.

Does faster flash translate into faster end-to-end LLM serving?

The paper tests the intuitive "SSD → HBF" upgrade at the full-system level.

Source: Fig. 3, abstract and introduction.

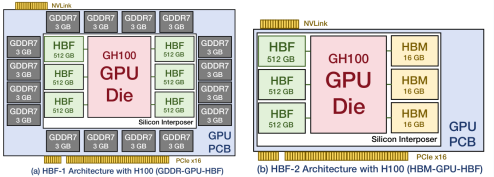
What Did the Paper Actually Evaluate?

METHOD / DROP-IN SUBSTITUTION

Keep the Mooncake-style runtime

Admission / retention / lookup / KV save & restore

Swap capacity-matched SSD for HBF-1 / HBF-2



(a) HBF-1(2028) (b) HBF-2(2030)

Figure 1: Modeled layouts based on the KAIST HBF roadmap.

Extended TokenSim

4 complete two-hour traces; 5 dense / MoE models
H100 / B200 profiles; thermal + endurance models

HBF-1 (2028): GDDR–GPU–HBF
Interposer sites go to HBF; near tier moves to off-interposer GDDR7.

HBF-2 (2030): HBM–GPU–HBF
HBM and HBF share sites; near-tier HBM capacity / bandwidth falls vs. SSD baseline.

HBF trades near-tier resources for package-local flash.

Source: Fig. 1, Tables 2 and 3(c), Sec. 6.1. Roadmap profiles + TokenSim + 3D-ICE + endurance model.

Three Conditions for HBF to Help

ORGANIZING PRINCIPLE

C1 — Critical Path

Read I/O must be a real serving bottleneck.

F1–F2: package tradeoff and limited base-die compute leverage.

C2 — Reuse

Useful reads must repay every HBF write.

F3–F4: cold write-heavy traffic and limited batching benefit.

C3 — Sustainability

High bandwidth must remain sustainable.

F5–F6: thermal throttling and flash write wear.

6 findings = 2 findings for each condition

A faster medium helps only if all three conditions hold.

Source: Sections 5–8.

Faster HBF Barely Improves E2E Latency

C1 FAILS / FLASH IS BARELY ON THE CRITICAL PATH

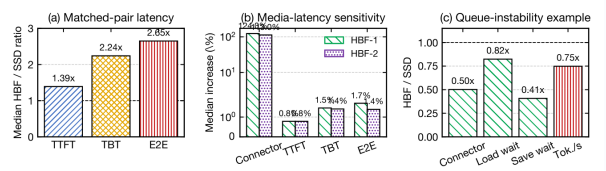


Figure 4: A faster connector, a slower system: HBF cuts transfer time yet raises end-to-end latency because the smaller near tier destabilizes the queue.

Mean E2E latency = **2–5.5× SSD baseline**
Less near-tier capacity / bandwidth
→ fewer admitted requests, more queueing

3.75× faster raw HBF latency
→ E2E only **0.75% / 0.88%** lower (HBF-1 / 2)
→ exposed secondary-tier I/O ≈ **1%**

The bottleneck is not flash-media latency.

Source: Fig. 4 and Sec. 6.1. HBF-1 vs. SSD24; HBF-2 vs. SSD12. H100/B200 simulation.

All Production Traces Send More Writes Than Reads to HBF

C2 FAILS / USEFUL READS DO NOT REPAY WRITES

Metric	traceA	traceB	coder	thinking
Write/read ratio	2.14×	4.90×	1.14×	2.20×
ρ = reads/write	0.47	0.20	0.88	0.45
Prefix hit rate	38.21%	52.46%	45.58%	44.12%

Table 8: flash-tier byte traffic over each full two-hour replay.

Hot, highly reused KV → retained in HBM / GDDR

Cold, low-reuse KV → evicted to HBF

Prefix reuse exists, but stays in the near tier.

Every trace writes more bytes than it reads from HBF.

The cache hierarchy gives HBF a write-heavy, low-reuse KV stream.

Source: Table 8 and Sec. 7.1. Production-trace replay, not HBF device measurements.

Write-Heavy KV Traffic Makes Peak HBF Bandwidth Unsustainable

C3 FAILS / THERMAL LIMITS AND WRITE WEAR

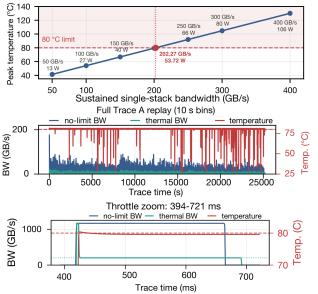


Figure 10: (a) Dynamic power and peak temperature vs. single-stack bandwidth. (b) Full replay: thermal control caps delivered bandwidth. (c) Throttle zoom: temperature stays near 80°C while bandwidth is capped.

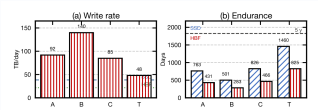


Figure 11: KV write volume and endurance budget. A–T denote traceA, traceB, coder, and thinking.

Short term: thermal throttling
202.27 GB/s at **80°C**
Delivered BW < peak (16-Hi TLC).

Long term: write endurance
Trace writes: **48–140 TB/day**
Modeled HBF lifetime ≈ **0.56× SSD**

Write-heavy KV reduces sustained bandwidth and consumes flash lifetime.

Source: Figs. 10–11 and Sec. 8. Model projections, not hardware lifetime measurements.

The Device Is Fine; the Drop-In Deployment Is Not

CONCLUSION / PLACEMENT REQUIREMENTS

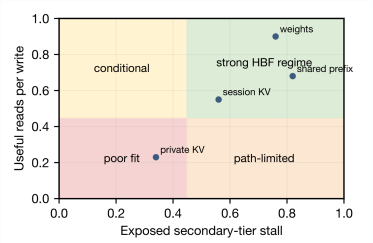


Figure 12: Placement map implied by the findings: exposed stall (C1) against reads-per-write (C2), with sustainable bandwidth (C3) shifting the boundary.

Poor HBF targets	**Better HBF targets**
Private KV	Model weights
Transient decode KV	Shared prefixes
Low-reuse objects	Read-mostly objects

Requirements for a future runtime

Critical-path-aware / reuse-aware
Write-budgeted / thermal-aware

HBF should not be treated as a faster SSD.

A conceptual design map from the findings, not an implemented runtime or calibrated boundary.

Source: Fig. 12 and Sections 9–10.

Sources Behind the HBF Evaluation

SYSTEM / WORKLOAD

TokenSim / HBF code

pku-lemonade research project
Open-source serving simulator
→ Serving, KV transfers, queueing and E2E latency

Qwen-Bailian traces

Alibaba / SJTU; production data
→ Request and reuse replay: traceA, traceB, coder, thinking

HBF DEFINITION

OCP HBF v0.7.0

Open Compute Project — **industry spec**
Contributors: Sandisk + SK hynix
→ Interface, page-write and lifecycle rules

KAIST HBF roadmap

Joungho Kim / KAIST TERA Lab
→ HBF-1 / HBF-2 organizations
Academic roadmap, not a product datasheet

HAVEN

Hsu et al.; academic architecture/model
→ NAND latency, energy, bandwidth, power

PHYSICAL LIMITS

3D-ICE

EPFL; open-source thermal tool
→ 16-Hi HBF temperature model
80°C policy: chosen by this paper.

KIOXIA CM7-V data

KIOXIA America; vendor datasheet
→ SSD baseline / rated endurance
HBF lifetime: authors' TLC budget + trace writes; not a wear-out test.

2028 / 2030 are roadmap generations, not guaranteed mass-production dates.

No physical HBF hardware is measured here: the full-stack model combines code, specifications, roadmaps, device models, production traces and thermal modeling.

Source: Paper Refs. [1], [9], [11], [13], [23], [24], [26]; Sections 2 and 4.